

Batteries alone are not sufficient to store energy from solar panels. There is a severe need for energy storage with management solutions. Concepts like smart technology and net metering will evolve in order to manage the energy reliably. The sector is expected to witness huge investment, large number of foreign and domestic players entering this domain and high level of awareness among customers.

Innovative storage mechanisms (such as flow batteries instead of lead-acid and lithium-ion technologies) and hybridising solar plants with either wind, pumped hydro or biomass can make load balancing more robust.

Powered by reliable backup

Growth of solar energy creates huge opportunities for the complete value chain and products required to install a complete solar setup, especially for solar inverters for more reliable backups. Inverters for solar systems are in great demand given the growing number of solar power installations and continuous growth ranging from turnkey projects to solar rooftops.

The Indian market is diversifying with the introduction of solar-related products and solar-integrated systems. R&D has become one of the major focus areas of the power backup systems market.

Solar technologies have evolved a lot since these first made their debut in the 1960s. While previously solar photovoltaics were seen as a thing of the future, today, technological breakthroughs have positioned the industry for huge growth. A series of new developments in solar photovoltaics technology also promise to contribute to the industry's success.

Consequently, the industry is witnessing a shift towards new technology and design in order to support more sophisticated appliances.

Smart inverters on the way

With technology being upgraded over time, solar power inverters have become smarter and offer a number of features that make these more efficient. Green UPS systems, which offer efficiency rates of 94.97 per cent, as against the usual 80.88 per cent efficiency units, are one of the latest trends that have emerged in the market.

Steps to boost solar growth

1. Faster and efficient implementation of renewable energy certificates
2. Carbon trading as a source of revenue
3. Selection of the right location for implementation of on-grid applications
4. Development of off-grid applications
5. Establishment of localised mini-grids, especially in regions that lack grid connectivity due to physical or financial barriers

Unique opportunities

Smartcities. The smartcity initiative being undertaken by the government is likely to entail significant investments in installing renewable energy capacity.

Railways. Indian Railways plans to leverage solar energy by installing solar power systems on rooftops of 500 trains as a pilot project.

Over time, solar power inverters have become smarter and offer a number of features such as remote monitoring and touchscreen technology that have increased their demand. Even micro-inverters have received significant importance in the global market. In terms of dimensions, compact inverters in central sizes that have lesser weight and lower losses are growing in popularity. Solar system developers such as engineering, procurement and construction companies are looking for inverter manufacturers that offer the latest technology and robust designs that will enable them to operate in different conditions.

The Indian power industry offers

both grid-tie and off-grid inverters. Considering unstable grids situations, off-grid solar inverters are more suitable as compared to grid-tie inverters, which have limited use in India currently, as these require solar as well as grid power.

Today, high-power-tracking systems such as maximum power point tracker based solar inverters are being used. This is the technology shift that is occurring from the typical pulse width modulation inverters. Companies are employing more of this tracking technology to extract the maximum from sunlight and convert it into usable power.

Another trend is hybrid inverters. There is lot of scope for development in DG-solar hybrid spaces as only a few companies are capable of manufacturing the same. Considering the net metering system in India, inverters with backup facility as well as the ability to interact with the grid will be in great demand.

Looking ahead

Overall, the next few years seems to be an era for the Indian solar sector. It could even be considered the onset of a golden era for solar energy sector if all good intentions of the government turn to reality.

However, lack of standardisation for off-grid systems makes designs and performance parameters difficult to achieve. The government needs to establish process guidelines and enforce stringent quality standards to ensure sustainability of off-grid systems.

Two big hindrances that impact the growth of solar sector in India are lack of proper knowledge among consumers and deficit of talented manpower who can sell/install solar.

Last, but not the least, sectorial skill development programmes need to be supported by both government and industry to ensure availability of huge manpower to support enhanced production and installation of solar systems. **EFY**